

# Bringing the Heat: from forecast to cost-optimal day-ahead bid for heat-pump homes

Team: Mona Alkhatay, [add all team member names] · E.ON challenge, RWTH hackathon

## Idea

E.ON must buy tomorrow's electricity for its customers by noon today, and every forecast error is settled later at a worse price. We treat the four levels as one chain of decisions: forecast the demand (L0), find who behaves alike (L1), price the forecast error in euros (L2), and choose the bid that minimises that cost (L3). The whole solution is four short Python scripts using pandas and scikit-learn only.

## Experimental design

Target: hourly grid imports of the whole portfolio (390 of 410 homes with enough data in both seasons), i.e. what E.ON actually buys; 15-min readings are summed to hours to match the weather and price resolution. Every forecast for day D is made at noon on D-1, so load inputs are at least 48 h old. Split is time-ordered: train Jan 2021 – Aug 2023, test Sep 2023 – Feb 2024 (last 24% of the data, no random split). Accuracy is  $nMAE = \text{mean absolute hourly error} \div \text{mean test load}$ ; ML results are averaged over 5 random seeds.

## Method

- **L0 Forecast.** Gradient-boosted trees (HistGradientBoostingRegressor) on load lags (48 h, 168 h, mean of D-2), hour, weekday, month, temperature and sunshine; baseline = same hour last week.
- **L1 Groups (unsupervised).** Each home is described by 52 features from its own meter data only: normalised winter and summer daily shapes, consumption level, heating sensitivity (kWh per °C below 15 °C) and a PV signature (correlation of midday import share with sunshine). k-medoids, k chosen by silhouette; k-means as a robustness check. The survey PV flag is never an input, only used to validate. One model per cluster, summed.
- **L2 Procurement metric.** Cost of error =  $\text{shortfall} \times 0.5 \cdot |p| + \text{surplus} \times 0.3 \cdot |p|$  with real hourly Swiss day-ahead prices p (Energy-Charts / SMARD); the +50% / -30% penalties are assumed and tested over 9 scenarios. Reported with bias, evening-peak error and worst-day error.
- **L3 Uncertainty and decision.** Because short costs more than long, the expected-cost-minimising bid is the  $\tau^* = 0.5 / (0.5 + 0.3) = 0.625$  quantile (newsvendor). Uncertainty = empirical quantiles of the model's relative errors over the previous 28 days, so only past information is used.

## Results (test period)

| Level | Main result                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L0    | Hourly nMAE 20.1% (naive) → 6.24% (model).                                                                                                                                                                                                                                                                                                                                                                                          |
| L1    | Silhouette clearly best at k = 2. Group A (70 homes) stops importing at midday in summer: 100% of its surveyed homes have PV and none of 108 non-PV homes is misplaced (recall 52%; the remaining PV owners import like non-PV homes). k-means agrees on 96% of homes. A model per group lowers nMAE to 5.96% (-4.6% relative, better in 5 of 5 runs), mainly in sunny months; the PV group is the uncertain one (13% vs 6% error). |
| L2    | Cost of error €92.6 → €24.9 per home per year (-73%). Yet the 4.6% kWh gain of clustering is worth only 0.2% in euros, and the better model flips across penalty scenarios: models must be chosen in euros.                                                                                                                                                                                                                         |
| L3    | Errors are widest around midday. The theoretical $\tau^* = 0.625$ coincides with the cheapest quantile in hindsight: -2.1% cost vs bidding the mean, -17% if shortfalls cost +100%. Full chain naive → L3 bid: €107 → €27 per home per year (from 30 Sep, after 28 days of error history).                                                                                                                                          |

## Design choices and originality

(1) PV owners are found without labels, from load shape alone, and validated against the survey. (2) Models are judged by what their errors cost, not by MAE alone, which reverses conclusions that kWh metrics suggest. (3) The bid is derived from theory (newsvendor) and confirmed empirically, using only information available at gate closure. (4) Everything is reproducible: four scripts, one run order, results written to CSV.

## Limitations

Measured weather stands in for a weather forecast (slightly optimistic, equal for all models); the data has no PV generation or export, so PV is seen only through imports; prices assume the homes are in Switzerland and the imbalance penalties are assumptions; one portfolio-wide quantile is used (per-cluster quantiles are the next step).